Main Gear Leg – General Data

1. General

Refer to Figure 1 for details.

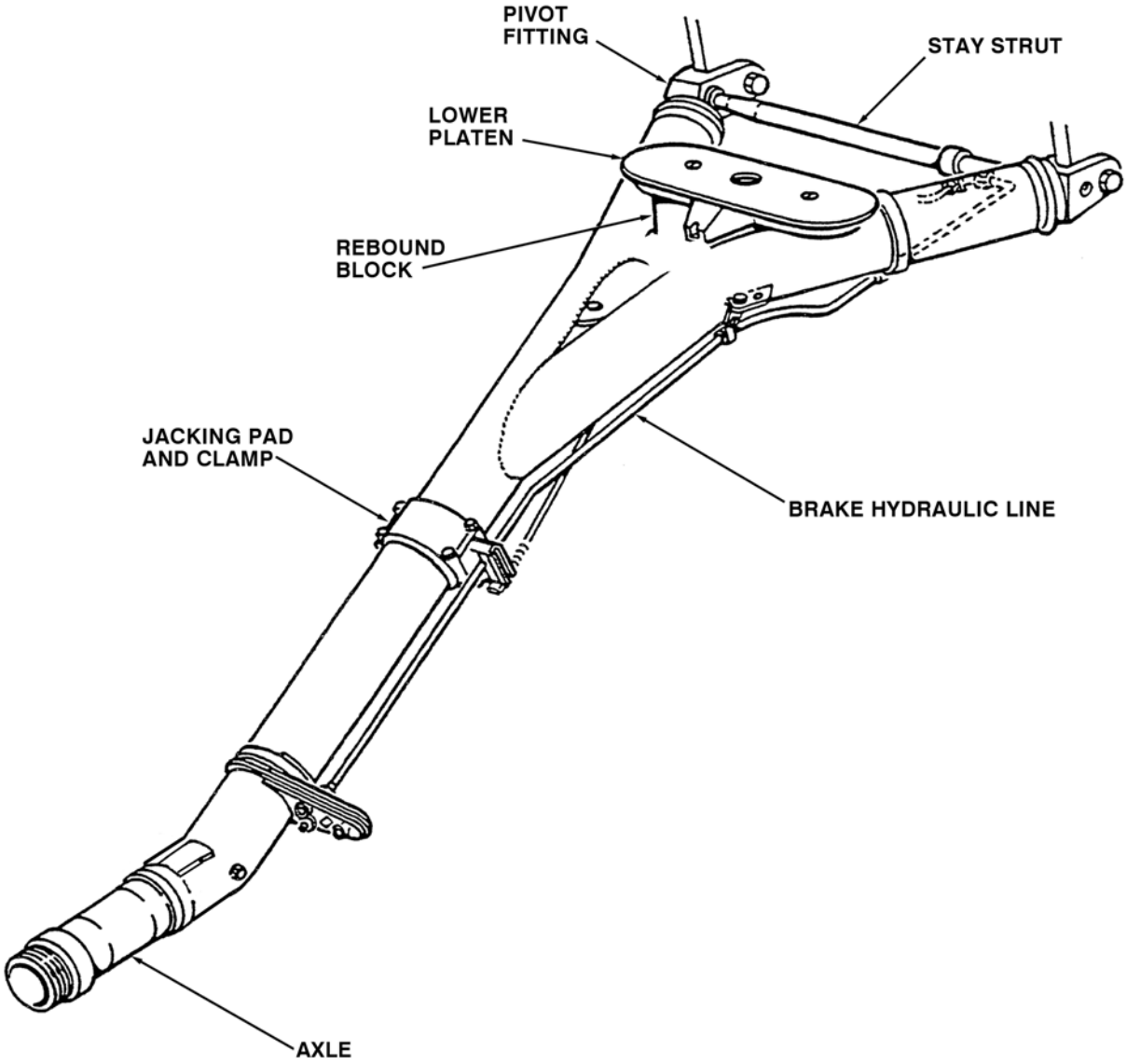
Each main gear leg consists of a Y-shaped tubular strut leg machining (C6UM1110) (Hy-Tuf Material), a stay strut and an axle. The Y-shaped strut is comprised of a front strut to which is welded a short rear strut. At the upper ends of both front and rear struts are pivot fittings which are welded to the struts. The two pivot fittings are interconnected by a stay strut. An axle is bolted to the lower end of the leg. Midway between the axle and the pivot fittings, a clamp type jacking pad is bonded to the strut. Four pads, each containing a threaded hole, are welded to the leg for attachment of the shock absorber lower platen. Each main gear leg is enclosed by front and rear section fairings.

A. Stay Strut

A stay strut is provided for each main gear leg and consists of a tube with two flanged end fittings. Each strut is positioned between the main gear leg pivot fittings and is shimmed equally at both ends to maintain the proper dimension between the pivot fittings. Screws and special washers secure the strut in position.

B. Main Gear Leg Fairings

Each main landing gear leg fairing is of fiberglass construction and is in two sections, front and rear, which are joined together by screws. The fairings are centered at the lower end of the leg by attachment to a split seal. The upper ends are attached to a frame which is bolted to two lugs on the leg. Contained in the frame is a roller which is free to turn in two nylon bearing cups. A blind which is attached to the fuselage structure rides over the roller and is held in tension by an elastic cord which is anchored to the front fairing. The blind provides protection for the shock absorber from dirt, water and debris entering the leg fairings during take-off and landing when the leg is in motion.



Main Gear Leg
Figure 1

2. Removal/Installation

Refer to Figure 2 for details.

A. Remove Main Gear Leg

1) Discharge brake system accumulator pressure by pumping brake pedals.

2) Remove main gear/fuselage fairing. Place container to receive spillage, and disconnect brake system flexible hose at elbow on stay strut. Cap open connection and line.

3) Jack aircraft until wheel is almost clear of ground (refer to 07-00-00, Lifting and Shoring – General Data), and remove shock absorber (refer to 32-10-00, Main Gear – General Data).

4) Remove pivot bolts at front and rear struts.

5) Disconnect brake system hydraulic line at lower end of leg. Cap open line. Remove wheel brake unit (refer to 32-40-31, Wheel Brakes Unit (Mod 6/2002) – General Data).

6) Remove leg fairings by removing attaching screws and unhooking blind elastic cord from inside of front fairing.

B. Install Main Gear Leg

1) Install wheel brake unit (refer to 32-40-31, Wheel Brakes Unit (Mod 6/2002) – General Data). Connect brake system hydraulic line at lower end of leg.

2) Position leg, insert pivot bolts and spacers at front and rear strut, using alignment pin SD12576–1 and acorn SD12576–3, and install washers and nuts. Tighten bolts to 200 to 400 pound–inches (dry thread) and install cotter pins.

3) Install shock absorber (refer to 32-10-00, Main Gear – General Data).

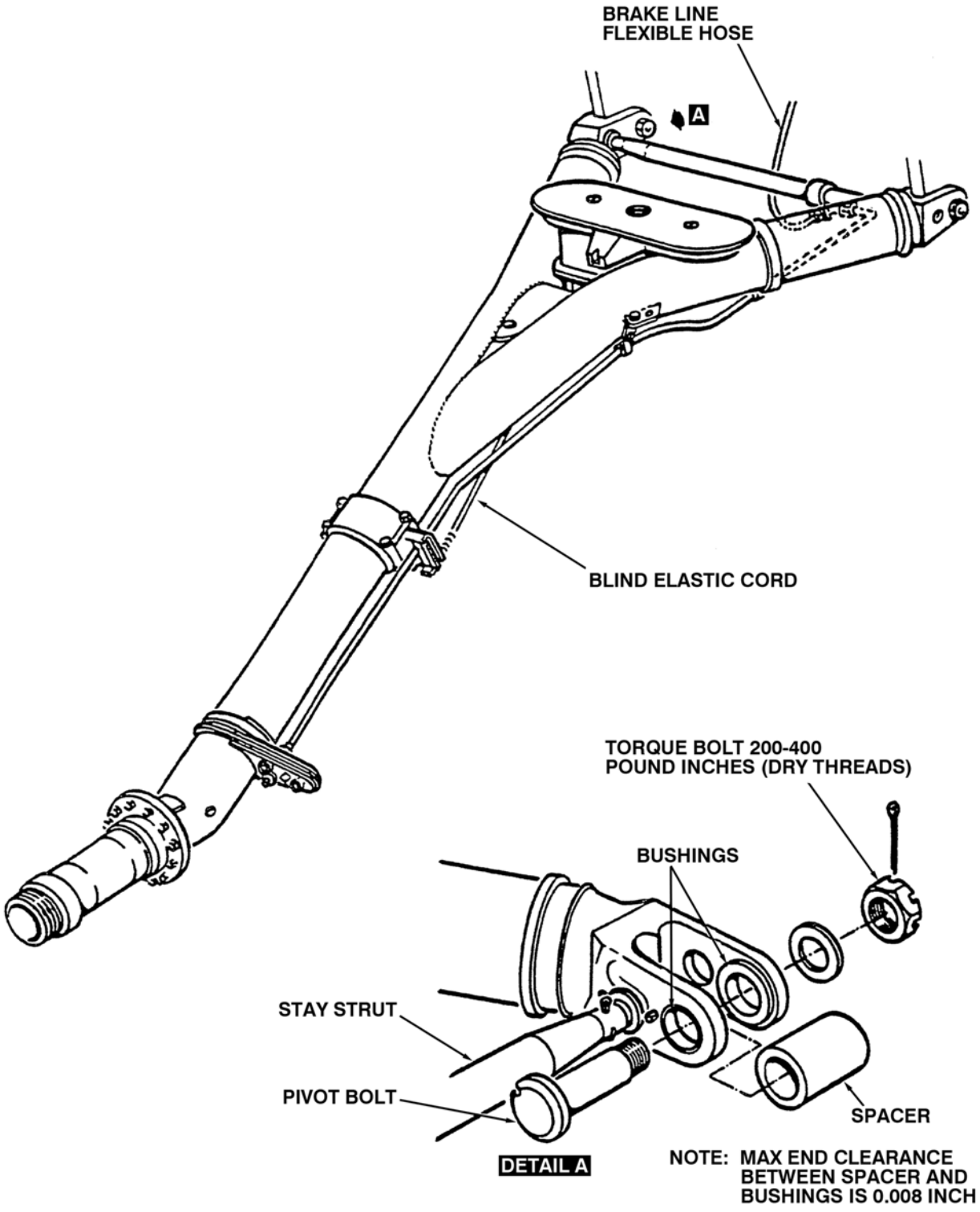
4) Connect brake system flexible hose at elbow on stay strut.

5) Bleed brake system (refer to 32-40-00, Wheels and Brakes (Mod 6/2002) – General Data, Para 2.).

6) Position leg fairings, secure attaching screws, and hook blind elastic cord to inside of fairing.

Note When a replacement main gear leg is installed, and fairing support frame is transferred from the unserviceable leg, the frame must be realigned with the two lugs on the leg by relocating the six washers on each bolt as required. The aircraft must be jacked up during this operation to ensure that leg will be in flight position (i.e., wheel clear of ground).

7) Install main gear fuselage fairing.



Main Gear Leg Installation
Figure 2

C. Gear Configuration Change

1) ESIS GH–3100 (Pre Mod 6/2170)

a) Change the DAU configuration plugs to match the gear configuration as per 31-44-00, Data Acquisition Unit (DAU) (Mod 6/2049) – General Data, Para 4.B.2).

b) Change the ESIS configuration plugs as per 31-20-21, Electronic Standby Instrument System (ESIS) (Mod S.O.O. 6209, Mod 6/2049 and Mod 6/2170) – General Data, Para 3.E.1).

Note If this is a first install for the ESIS configuration module, a compass swing must be done to ensure the ESIS is calibrated correctly.

c) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

Note If file identification is required for the MAU option files - aircraft Serial number specific, and ADAHRS certification files - gear configuration type specific, contact Viking Air Ltd. Technical Support.

d) Load the MAU option file to match gear configuration. (Refer to 31-43-00, Modular Avionics Unit (MAU) (Mod 6/2049) – General Data).

e) Load ADAHRS certification file to match gear configuration. (Refer to 34-23-00, Air Data, Attitude and Heading Reference System (ADAHRS) (KSG 7200) – General Data).

f) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

g) Do a configuration check, refer to Para 3.A.

2) ESIS GH–3900.2 (Post Mod 6/2170)

a) Change the DAU configuration plugs to match the gear configuration as per 31-44-00, Data Acquisition Unit (DAU) (Mod 6/2049) – General Data, Para 4.B.2).

b) Load GH–3900.2 configuration file to match gear configuration. (Refer to 31-20-21, Electronic Standby Instrument System (ESIS) (Mod S.O.O. 6209, Mod 6/2049 and Mod 6/2170) – General Data, Para 3.E.2)).

c) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

Note If file identification is required for the MAU option files - aircraft Serial number specific, and ADAHRS certification files - gear configuration type specific, contact Viking Air Ltd. Technical Support.

d) Load the MAU option file to match gear configuration. (Refer to 31-43-00, Modular Avionics Unit (MAU) (Mod 6/2049) – General Data).

e) Load ADAHRS certification file to match gear configuration. (Refer to 34-23-00, Air Data, Attitude and Heading Reference System (ADAHRS) (KSG 7200) – General Data).

f) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

g) Do a configuration check, refer to Para 3.A.

3. Adjustment/Test

A. Configuration Check

1) Ensure the aircraft electrical system is OFF.

2) Open the DAU 1A (E5), DAU 2A (F5) (left bus), DAU 1B (E6), DAU 2B (F6) (right bus) circuit breakers.

3) Gain access to the ceiling compartments aft of bulkhead 240.

4) Confirm that the configuration plugs match the current gear configuration, VSP18–XXXX.

Note The VSP18–XXXX part number is unique to each aircraft's configuration options. Contact Viking Technical Support for configuration option confirmation.

5) Close the DAU 1A (E5), DAU 2A (F5) (left bus), DAU 1B (E6), DAU 2B (F6) (right bus) circuit breakers.

6) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

7) Confirm CAS Advisory message with current gear configuration in table below:

CAS Advisory Message	Gear Type
Gear Wheels	Standard Wheel
Gear Amphibious	Float 1 Amphibious
Gear Wipline Floats	Float 2 Wipline Floats
Gear IFG	Intermediate Flotation Gear
Gear Wheel-Ski	Wheel-Ski

8) If there is CONFIG_MISCOMPARE Warning CAS messages for the gear type. Refer to Para 6.

9) Confirm ESIS display matches the current gear configuration as per 31-20-21, Electronic Standby Instrument System (ESIS) (Mod S.O.O. 6209, Mod 6/2049 and Mod 6/2170) – General Data), Para 3.F.

10) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

4. Inspection/Check

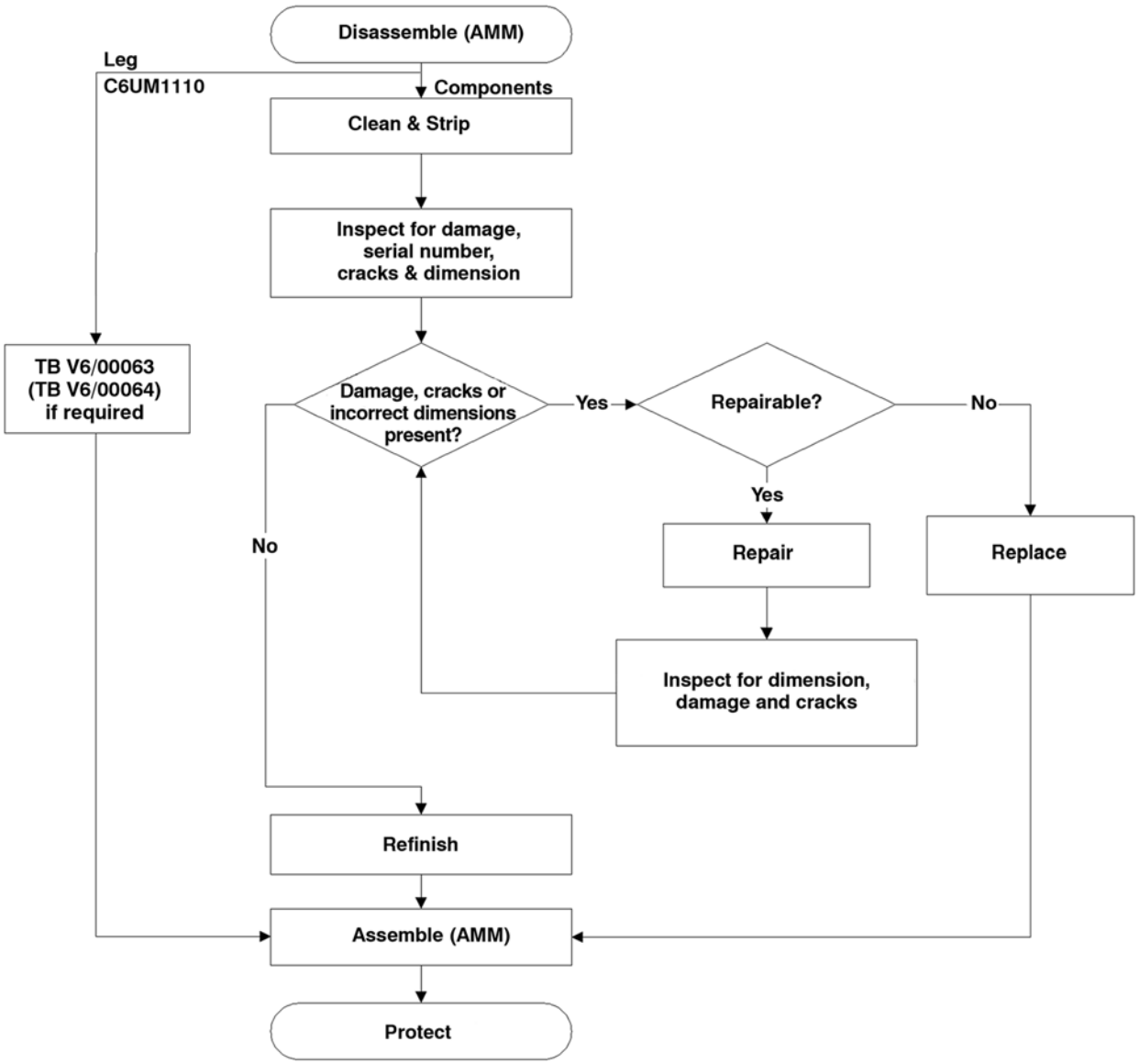
A. General

The general sequence for inspection and repair of the DHC-6 Main Landing Gear (refer to Figure 1) consists of the following:

1 The inspection and repair of the Main Landing gear components, in accordance with Aircraft Maintenance Manuals.

2 The inspection and repair of the DHC-6 Main Landing Gear Leg Machining (C6UM1110) (Hy-Tuf Material) and the pivot fittings, in accordance with TB V6/00063 for inspection, and (if required) TB V6/00064 for repair.

3 Protection of the MLG for short term storage (up to 6 months) in accordance with Para 5.B should proceed immediately after overhaul. Either the leg machining (C6UM1110) alone or the assembled MLG leg can be protected. If time will elapse between overhaul and assembly, protect the leg machining (C6UM1110) as per Para 5.B in the interim.


Disassemble (AMM)
Leg
C6UM1110
Components
Clean & Strip
Inspect for damage, serial number, cracks & dimension
TB V6/00063 (TB V6/00064) if required
Damage, cracks or incorrect dimensions present?
Yes
Repairable?
No
Yes
No
Repair
Replace
Inspect for dimension, damage and cracks
Refinish
Assemble (AMM)
Protect

1) Equipment and Materials

Facilities shall be equipped, in addition to standard shop tools, with the equipment listed below. Where applicable, equipment shall carry a current calibration and/or certification certificate and date.

Materials required for the removal, cleaning, inspection, finishing, assembly, and packing of the main landing gear leg assembly are provided below.

Note Use of qualified materials, where applicable, is mandatory.

a) Equipment and Tooling

- Compressed air supply, clean and dry
- Hot and cold water supplies
- Vapour degreasing equipment conforming to ASTM D3698
- Micrometer calibrated to ±0.0001 inch or better
- Telescoping gauge set
- Magnetic particle inspection equipment per ASTM E1444
- Fluorescent penetrant inspection equipment per ASTM E1417
- Heat lamp
- Paint spray/application equipment
- Hydraulic pressurization test equipment
- Coating thickness gauge
- Torque wrench
- Lint-free cloths (MIL–DTL–2467/C)
- Lint-free cotton gloves
- Solvent resistant gloves
- Stir sticks
- Tack rags
- Steel wire wheel
- Steel wool

b) Materials

- Abrasive paper, aluminum oxide, 180 – 220 grit (Commercial)
- Adhesive, RTV730 (Dow Corning)
- Adhesive, EC1357 (3M)
- Celastic tape, 0.050 inch thick recommended (Commercial)
- Conversion coating, Alodine 1200S or 1201 (MIL–DTL–81706)
- Corrosion preventive compound, Grade 1 (MIL–PRF–16173 Grade 1)
- Corrosion preventive compound, Grade 2 (MIL–PRF–16173 Grade 2)
- Degreasing solvent (MIL–PRF–680, Type I)
- Developer, fluorescent penetrant inspection (QPL–AMS–2644)
- Diestone DLS (Socomore)
- Diestone HFP (Socomore)
- DS108 (Dysol)
- Enamel, polyurethane (MIL–PRF–85285, Type I, Class H)
- Epoxy primer (MIL–PRF–23377)
- Fluorescent dye (QPL–AMS–2644)
- Grease (MIL–PRF–23827)
- Hydraulic fluid (MIL–H–5606)
- Kraft paper, oil-resistant and water repellent (Commercial)
- Loctite ‘H’ Sealant (ASTM D5363, AN0161)
- Loctite ‘N’ 7649 Primer (ASTM D5363, AN0321)
- Lubricant (MIL–L–7870)
- Magnetic wet particles, fluorescent (AMS 3045 or AMS 3044)
- Masking tape (Commercial)
- Methyl Ethyl Ketone (MEK) (ASTM D740)

- Non-acid paint stripper (Commercial grade)
- Sodium hydroxide (Commercial grade)
- Stoddard solvent (MIL–PRF–680)
- Suspension for magnetic particles (DOD–F–87935 or AMS 2641, Type I)
- Toluene (A–A–59107)
- Trichloroethane (MIL–T–81533)
- Trichloroethylene (ASTM D4080)
- EnSolv–5408 (stabilized n-Propyl Bromide based solvent), Enviro Tech
- EnTron–Aero 150 (stabilized n-Propyl Bromide based solvent)
- Leksol (stabilized n-Propyl Bromide based solvent), Amity International
- Varsol (ExxonMobil Chemical)
- VCI paper (MIL–PRF–3420 or meet the requirements of NACE TM0208)

c) Standards

- ASTM D3698, Standard Practice for Solvent Vapour Degreasing Operations
- ASTM E1417, Standard Practice for Liquid Penetrant Testing
- ASTM E1444, Standard Practice for Magnetic Particle Testing
- ISO/IEC 17025, General Requirements for the Competence of Testing and Calibration Laboratories

2) Standard Cleaning Methods

a) Solvent Cleaning External Surfaces

1 Use solvent applicable to material and cleaning purpose as listed in Table 1 — Solvent Use by Purpose and Material or Finish.

2 Apply solvent to a clean, dry, lint-free cloth and wipe the surface.

3 Dry with a clean cloth before solvent evaporates using a clean part of the cloth for each pass.

b) Solvent Cleaning Internal Surfaces

1 Plug one end of the part to be washed, as applicable.

2 Fill with solvent applicable to material and cleaning purpose as listed in Table 1 — Solvent Use by Purpose and Material or Finish.

3 Plug other part opening(s).

4 Hold the part horizontally and revolve part to ensure solvent reaches all internal surfaces.

5 Drain.

6 Repeat Para 4.A.2).b).2 through Para 4.A.2).b).5 until all dirt, and/or coatings and/or residue are removed.

Table 1 Solvent Use by Purpose and Material or Finish

Purpose	Material or Finish	Applicable Solvent(s)
Removal of oil, dirt, dust, etc.	Primed surfaces	DS108, Diestone DLS, Diestone HFP
	Painted parts other than axle, leg machining	DS108, Degreasing solvent (MIL–PRF–680, Type I)
	Painted axle, leg machining	DS108, MEK
	Plated parts	DS108, MEK
	Bare metal	DS108, Diestone DLS, Diestone HFP, MEK
	Bushings	DS108
Removal of uncured corrosion preventive compound or uncured adhesive	All metal parts	DS108, Diestone DLS, Diestone HFP, MEK
Solvent wash removal of cured corrosion preventive compound	Steel parts	MEK, trichloroethylene, trichloroethane, or equivalent
Preparation for installation	Fasteners	DS108 or MEK
Preparation for hydraulic test	Brake tube	Stoddard solvent or Varsol
Cleaning after hydraulic test	Brake tube	Stoddard solvent
Preparation for painting	Fully cured primed surfaces	DS108, Diestone DLS, Diestone HFP
	Partially cured primed surfaces	DS108, Diestone DLS, Diestone HFP, MEK
Preparation for application Celastic tape	Primed or painted	DS108
Preparation for re-anodizing	Aluminum	DS108, Diestone DLS, Diestone HFP, MEK
Vapour degreasing	All metal parts	Trichloroethylene or equivalent

c) Vapour Degreasing

Note Vapour degreasing is the preferred method for cleaning internal surfaces.

1 Mask bushings and bearings.

2 Vapour degrease per ASTM D3698 using trichloroethylene, methyl chloroform, or equivalent.

3 Rack or arrange parts vertically.

4 Vertical rate of entry not to exceed 11 ft/min.

5 After degreasing, raise part(s) and drain until no solvent remains on the part(s).

d) Mechanical Cleaning

1 Surfaces to be mechanically cleaned must be free of oil, grease, and other shop contaminants.

2 Use a steel wire wheel, steel wool, or abrasive paper (180 – 220 grit) to remove scale, oxide, or rust.

Note Cleaning tools shall be clean, free of contamination and/or particles of other metal types.

3) Standard Temporary Protection Methods

a) Temporary Protection of Exposed Steel Surfaces

1 If subsequent operations or inspections are not to occur immediately, temporarily protect exposed steel surfaces with oil conforming to MIL–L–7870, or with MIL–PRF–16173, or equivalent. Ensure parts have a continuous film of oil on all internal and external surfaces, as applicable. Alternatively, wrap part(s) in VCI paper (MIL–PRF–3420 or meeting the requirements of NACE TM0208).

2 Do not coat with oil, parts that have been cadmium or chrome plated, primed, or painted.

b) Temporary Protection of All Parts

1 If subsequent operations or inspections are not to occur immediately, wrap part(s) coated with MIL–L–7870 or MIL–PRF–16173 in kraft paper and secure with masking tape.

4) Standard Inspection Methods

a) Dimension Measurements

1 Measure dimensions using calibrated calipers with a tolerance of ±0.0001 inch.

b) Magnetic Particle Inspection

1 Carry out Magnetic Particle Inspection per ASTM E1444 using Fluorescent wet method with DC current.

2 Prior to inspection, ensure all paints and coatings (metallic and otherwise) have been removed and that surfaces are free of oil, dust, and shop contaminants.

3 Use best practices. Refer to Table 2 — Parameters for Magnetic Particle Inspection.

Note Values are provided for reference only. Parameters may vary depending on equipment and operator experience.

Table 2 Parameters for Magnetic Particle Inspection

Parameter	Value
Magnetizing current	DC
Field Strength	30 - 60 Gauss (2.4 - 4.8 kA/m)
Magnetization method	2 shots 0.5s - 1.0s 2 directions 90° apart
Particle	Wet fluorescent per AMS 3045 or AMS 3044
Suspension vehicle	DOD–F–87935 or AMS 2641, Type I
Particle concentration	0.1 - 0.4% by volume
Light source	Black light (UV-A), ≥1200 μW/cm^2 @ 15 inches

Table 2 Parameters for Magnetic Particle Inspection (continued)

Parameter	Value
Background illumination	≤2 ft-candles (20 lux)
Maximum residual magnetic field intensity	3 Gauss (0.24 kA/m)

4 Inspect exterior and interior surfaces. For interior surfaces, use a boroscope with the appropriate light source.

5 Inspect the whole of the part, paying particular attention to stress concentrations (e.g. holes, abrupt changes in shape), and weld areas.

6 De-magnetize.

7 Remove all magnetic particle solution by solvent cleaning, in accordance with Para 4.A.2).a) and Para 4.A.2).b).

8 Temporarily protect, in accordance with Para 4.A.3).

c) Fluorescent Penetrant Inspection

1 Carry out Fluorescent Penetrant Inspection per ASTM E1417 for all non-magnetic parts.

2 Prior to inspection, ensure all paints and coatings have been removed and that surfaces are free of oil, dust, and shop contaminants.

3 Use Type I (fluorescent dye), Method A (water washable), Sensitivity Level 3 (high) penetrant system conforming to QPL–AMS–2644.

4 Use a suitable developer, conforming to QPL–AMS–2644.

5 Use best practices.

6 Remove all inspection materials by solvent cleaning, in accordance with Para 4.A.2).a) and Para 4.A.2).b).

7 Temporarily protect in accordance with Para 4.A.3).

B. Inspection of Main Landing Gear Components

1) General

a) This section describes the minimum requirements for inspection of the main landing gear assembly components.

b) Fits and clearances for close parts must meet the requirements of Para 4.B.8) and PSM 1-6-3, Table 1, and Figure 1.

c) All parts retained for reassembly must be inspected for cracks. Standard parts (bolts, washers, pins, nuts, etc.), should be discarded and replaced with new parts.

2) Remove Main Landing Gear Leg from aircraft (refer to Para 2.A).

3) Remove Main Landing Gear Components C6U1103–3/–4 (refer to Figure 3 and Figure 4 for details) as follows:

a) Remove Brake Tube Assembly

1 Remove nut (1), tube clamp (2), and spacer (3) from clamp (38) at top of leg assembly.

2 Remove nut (4), washer (5), screw (6), clamp (7), nut (8), washer (9), angle (10), and bolt (11) from machined leg pad.

3 Remove bolt (12), washer (13), channel (14), and fairlead (15) from bracket at jacking pad.

4 Disconnect the tube from the union at rib, and from elbow at stay assembly. Remove the tube assembly (16).

Note If the leg assembly is fitted for skis, remove tube assembly and electrical wiring for ski position indication.

5 Remove nut (17), washer (18) and elbow (19) from stay assembly at top of leg assembly.

6 Remove nut (20) and union (21), two plugs (22), and two packings (23) from rib (25) at bottom of leg assembly.

b) Remove Lower Leg Rib

1 Release bond seal (26) from leg and rib (25), using toluene.

2 Loosen and remove clamp (24) and remove rib (25).

3 Remove Celastic tape (27) from rib (25).

Note Celastic tape will not be present for leg assemblies fitted for skis. This tape covers the hole provided in the rib for ski position electrical wiring.

c) Remove Axle

1 Remove nut (28), washer (29), and bolt (30).

2 Remove axle (31).

d) Remove Jacking Pad

1 Remove four nuts (32), four washers (33), two bolts (34), two bolts (34A), and remove bracket assembly (35) from clamp (36) and jacking pad (37).

2 Using MEK or DS108, loosen bond of clamp and jacking pad to leg, and remove clamp (36) and jacking pad (37).

e) Remove Upper Clamp

1 Using MEK or DS108, loosen and remove clamp (38) above lower platen.

2 Remove rubber sheet (39).

f) Remove and Disassemble Platen and Rebound Assemblies

1 Remove two nuts (40), two washers (41), two screws (42), two spigots (43), scuff plate (44) and liner (45) from platen assembly (48).

2 Remove four twelve-point bolts (46) and four washers (47).

3 Remove platen assembly (48), including rebound rubber assembly (55).

4 Remove four screw locking inserts (49) from leg machining pads.

5 Remove nut (50), washer (51), bolt (52), two bolts (53), and two washers (54).

6 Remove rebound rubber assembly (55) from platen assembly (48).

7 Remove two screw locking inserts (56) from platen assembly.

Note Do not remove four bushings (57) from platen assembly. Bushings need only be removed if repair is required (refer to Para 4.B.8)).

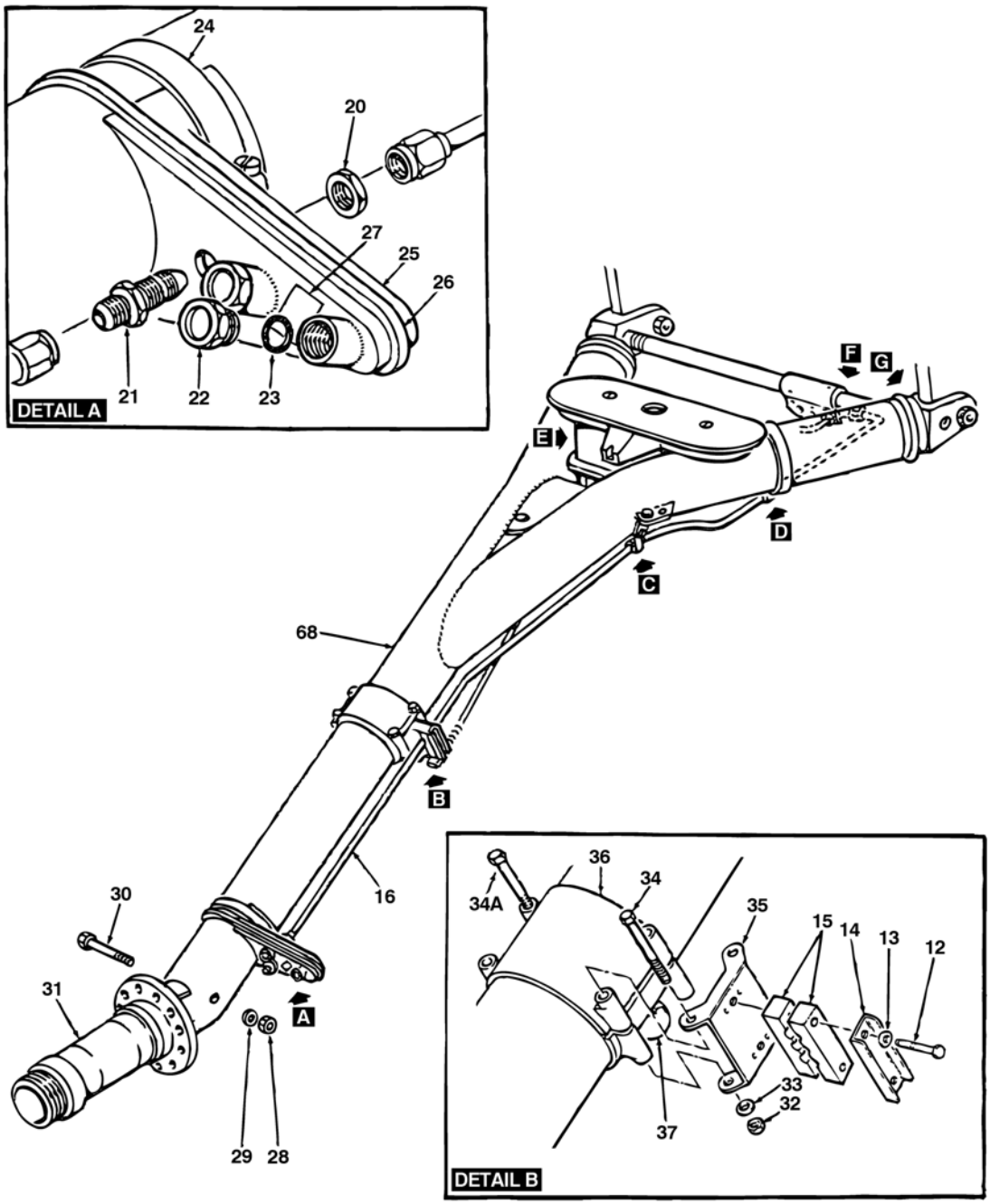
g) Remove Stay Assembly and Pivot Fitting Bushings

1 Remove spring pin (58) at rear stay pivot fitting.

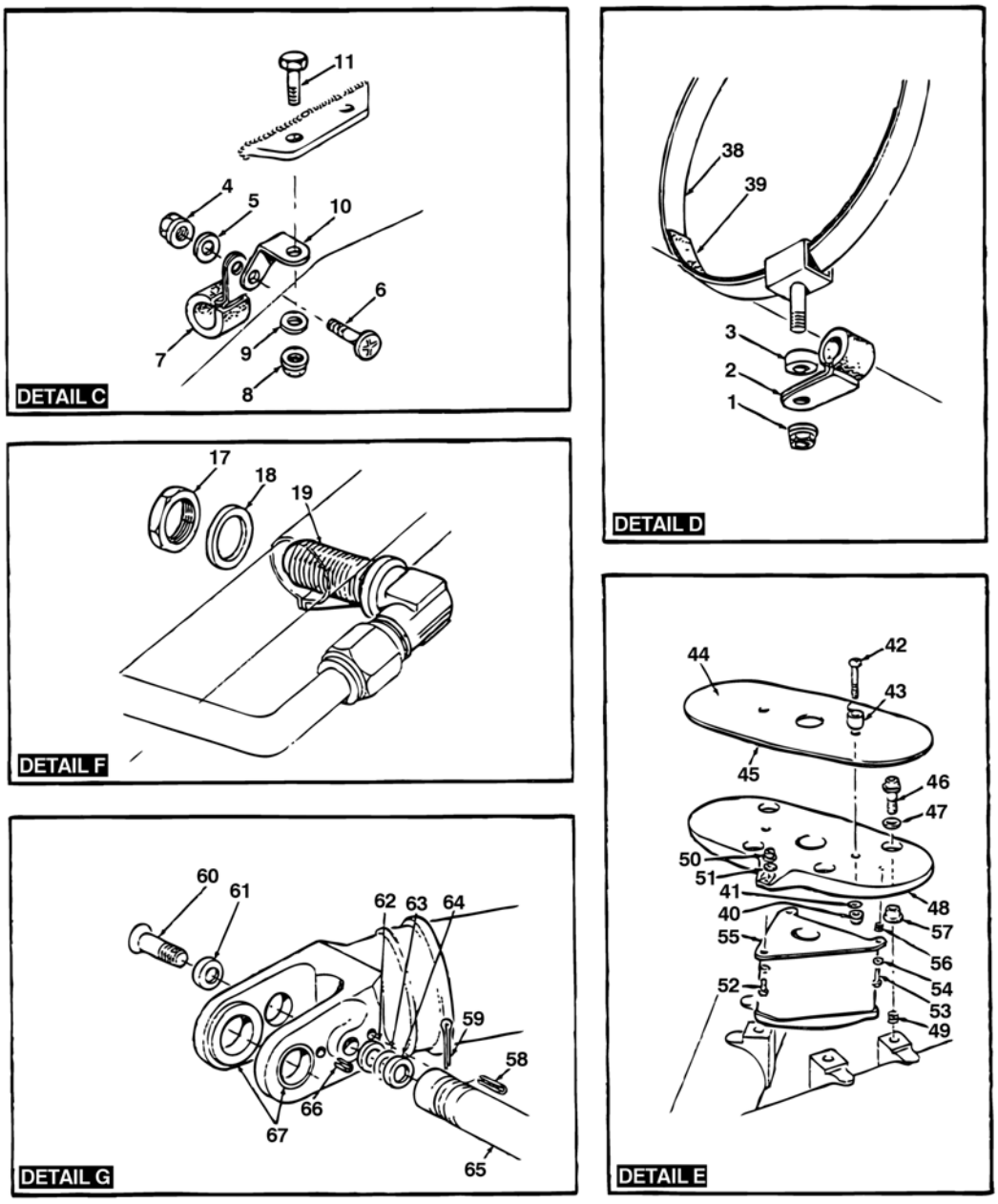
2 At front and rear pivot fittings, remove cotter pin (59), bolt (60), washer (61), washer (62) (if fitted), shim (63), washer (64), and remove stay assembly (65).

3 At front and rear pivot fittings, remove spring pin (66).

4 Remove four bushings (67) from pivot fittings.



Main Landing Gear C6U1103–3/–4 with Labeled Sub-Assemblies and Exploded Views
Figure 3



Additional Exploded Views for Main Landing Gear C6U1103–3/–4
Figure 4

4) Main landing gear leg machining (C6UM1110) (Hy-Tuf Material) and the pivot fittings are to be inspected at an appropriately rated inspection facility, in accordance with TB V6/00063, and, if required, repaired at an appropriately rated repair facility, in accordance with TB V6/00064.

5) Clean and Strip Main Landing Gear Components

   a) Cleaning on Receipt

      1 Remove dirt and surface contaminants by solvent cleaning parts, in accordance with Para 4.A.2).a) and Para 4.A.2).b), as applicable.

   b) Remove Corrosion Preventative Compound from Internal Surfaces

      1 Remove internal protective treatment by vapour degreasing, in accordance with Para 4.A.2).c), or by solvent washing, in accordance with Para 4.A.2).b).

      2 Visual inspect to ensure all traces of coating are removed. If not, repeat Para 4.B.5).b).1.

      3 Temporarily protect interior surface, in accordance with Para 4.A.3).

   c) Remove Paint from External Surfaces

      1 Clean exterior surfaces by vapour degreasing, in accordance with Para 4.A.2).c) (preferred) or by solvent cleaning, in accordance with Para 4.A.1).

      2 Remove external paint using a non-acid type paint stripper (Turco 5469, Cee–Bee R–256, Trebor 4500–PR55, Stripit 772, Oakite 157, Ardox 20, or equivalent).

         Note Paint must be removed from all parts to be inspected by Magnetic Particle Inspection or Fluorescent Penetrant Inspection.

      3 Immediately after stripping, solvent clean the external surface in accordance with Para 4.A.2).b).

      4 If subsequent operations are not to occur immediately, temporarily protect, in accordance with Para 4.A.3).

6) Inspect Main Landing Gear Components – General

   a) Visually inspect all part surfaces (outside and inside) for corrosion, plating defects, evidence of fatigue, scratches, cracks, and salt entrapment.

7) Inspect Main Landing Gear Components – Crack Detection

   a) Parts Compromised of Magnetic Metal Material

      1 Ensure that all paints and protective coatings have been removed and that parts are free of oil, dust, dirt and any other shop contaminants, in accordance with Para 4.B.5).a).

      2 Plated parts having electroplated finishes of 0.0008 inch or less, may be inspected without removal of the plating.

      3 Magnetic Particle Inspect, in accordance with Para 4.A.4).b).

      4 Parts having cracks in the parent material shall be replaced.

      5 Parts showing cracks in the plating can be repaired (refer to PSM 1-6-3, Section 32-10-00, Para 3.C.), or replaced.

b) Parts Compromised of Non-Magnetic Metal Material

1 Ensure that all paints and protective coatings are removed and that parts are free of oil, dust, dirt and any other shop contaminants in accordance with Para 4.B.5).a).

2 Fluorescent Penetrant Inspect, in accordance with Para 4.A.4).c).

3 Parts having cracks shall be replaced.

8) Inspect Main Landing Gear Components – Specific Components

a) Dimensional Inspections

1 Do all dimensional measurements in accordance with Para 4.A.4).a).

2 Many of the main landing gear component parts are close tolerance parts and are subject to wear.

3 Dimensions for fits and clearances are provided in PSM 1-6-3, 32-10-00, Table 1, and Figure 1. The table lists the “Original Manufacturing Limits” and “In-Service Wear Limits” for the corresponding part.

b) Main Landing Gear Axle (C6UM1134)

1 Visually inspect the plating for wear, scratches, and other damage or defects.

2 Visually inspect holes and diameters for wear, eccentricity, and other defects.

3 Visually inspect threads for wear and other defects.

4 Measure and record outer diameter of the axle bearing bands (refer to PSM 1-6-3, Section 32-10-00, Figure 1, Item A).

5 Measure and record axle bolt hole inner diameter (both holes), and axle bolt external diameter (refer to PSM 1-6-3, Section 32-10-00, Figure 1, Item B). Calculate and record clearance.

6 Axle not meeting outer diameter or plating requirements can be repaired (refer to PSM 1-6-3, Section 32-10-00, Para 3.C.).

7 Repair bearing surfaces in accordance with PSM 1-6-3RS, Chapter 32, RD6–32–016.

c) Ski Axle and Plug (C6US1015 and C6US1016)

1 Visually inspect the plating for wear, scratches, and other damage or defects.

2 Visually inspect holes and diameters for wear, eccentricity, and other defects.

3 Visually inspect threads for wear and other defects.

4 Measure and record outer diameter of the axle bearing bands (refer to PSM 1-6-3, Section 32-10-00, Figure 1, Item A).

5 Measure and record axle bolt hole inner diameter (both holes), and axle bolt external diameter (refer to PSM 1-6-3, Section 32-10-00, Figure 1, Item B). Calculate and record clearance.

6 Measure and record ski axle spline minor, major, and pitch diameters (refer to PSM 1-6-3, Section 32-10-00, Figure 1, Item E).

7 Ski axle and plug not meeting outer diameter or plating requirements can be repaired (refer to PSM 1-6-3, Section 32-10-00, Para 3.C.).

d) Lower Platen Assembly

1 Measure and record inner diameter of all four bushings (refer to PSM 1-6-3, Section 32-10-00, Figure 1, Item D).

2 Check that locking inserts are secure.

3 Visually inspect threaded holes for thread stripping, wear, or other defects.

4 Lower platen bushings and locking inserts not meeting dimensional tolerance must be replaced (refer to PSM 1-6-3RS, Chapter 32, RD6–32–016 for bushing repair limits or PSM 1-6-3, Section 32-10-00, Para 3.E.).

5 Repair lower platen holes and threads in accordance with PSM 1-6-3, Section 32-10-00, Para 3.E.

Note If the part cannot meet the requirements of PSM 1-6-3RS, Chapter 32, RD6–32–016 for bushing repair limits or PSM 1-6-3, Section 32-10-00, Table 1 upon repair, replace.

e) Brake Tube Assembly Inspection and Test

1 Visually inspect for holes or evidence of wear.

2 Flush line with Stoddard solvent or Varsol.

3 Ensure tubes inside and outside are dry.

4 Using hydraulic fluid conforming to MIL–H–5606, pressure test at 1.5 times the system operating pressure.

5 Ensure brake tube does not burst, leak, or otherwise deform under the test pressure. Replace brake tubes not meeting acceptance criteria.

6 Drip drain brake tubes and cap the ends.

7 Remove excess hydraulic fluid from the outside by solvent cleaning in accordance with Para 4.A.2).a).

9) Finish Axle Internal Surface as follows:

a) Prepare Internal Surfaces

1 Solvent clean internal surfaces, in accordance with Para 4.A.2).b) or vapour degrease in accordance with Para 4.A.2).c).

2 Visually inspect the internal surfaces to ensure all surfaces are free of corrosion products or other forms of contamination.

b) Apply Corrosion Preventive Compound to Main Landing Gear Axle (C6UM1134)

1 Plug one end of the axle using a suitable adhesive tape.

2 Completely fill the axle with undiluted corrosion preventive compound, conforming to MIL–PRF–16173, Grade 1.

3 Plug the open end of the axle using a suitable adhesive tape.

4 With the axle in the horizontal position, rotate the sealed axle 360° about the longitudinal axis several times to ensure all of the internal surface is covered. Let stand 5 minutes.

5 Drain and let stand for 16 hours.

6 Repeat protective treatment, Para 4.B.9).b).2 to Para 4.B.9).b).4.

7 Drain and let stand for a minimum of 8 hours.

8 Inspect all internal surfaces to ensure complete coverage of coating, if necessary, repeat Para 4.B.9).b).1 through Para 4.B.9).b).7 to obtain a satisfactory coating.

9 Remove excess compound from exterior surfaces by solvent cleaning, in accordance with Para 4.A.2).a).

c) Apply Corrosion Preventive Compound to Main Landing Gear Ski Axle (C6US1015)

1 Mask the splines and plug one end of the axle using a suitable adhesive tape.

2 Fill the axle with undiluted corrosion preventive compound, conforming to MIL–PRF–16173, Grade 1, taking care to not get any compound on the splines. Fill slowly to prevent air bubble formation. Let stand 5 minutes.

3 Puncture the plugged end and allow compound to slowly drain.

4 Position the part vertically and let stand for 16 hours.

5 Repeat protective treatment, Para 4.B.9).c).2 to Para 4.B.9).c).3.

6 Position the part vertically and let stand for a minimum of 8 hours.

7 Inspect all internal surfaces to ensure complete coverage of coating. If necessary, repeat Para 4.B.9).c).1 through Para 4.B.9).c).6 to obtain a satisfactory coating.

8 Brush a thin, uniform coating of undiluted corrosion preventive compound, conforming to MIL–PRF–16173, Grade 2, onto the ski axle splines, overlapping slightly with the Grade 1 coating. Inspect to ensure complete coverage.

9 Remove excess compound from exterior surfaces by solvent cleaning, in accordance with Para 4.A.2).a).

10) On parts other than leg machining or axle, restore damaged external finish as necessary using fluid resistant epoxy primer to MIL–PRF–23377 and grey polyurethane enamel to MIL–PRF–85285, Type I, Class H. As an alternative, zinc chromate primer may be used in place of epoxy primer. The use of etch primer is not recommended.

11) Install Main Landing Gear Components C6U1103–3/–4 (refer to Figure 3 and Figure 4 for details) as follows:

Note Ensure all parts are clean and free of defects prior to assembly. Only newly inspected and repaired, or replaced, main landing gear components are to be installed on newly inspected and, if required, repaired leg machining (C6UM1110), or new leg machining.

a) Install Stay Assembly

1 Ensure bushings (67) (or oversize bushings) are secure in pivot fittings.

2 Apply corrosion protective compound (Grade 2) to holes for spring pins, and install spring pins (66) at front and rear pivot fittings, and spring pin (58) at rear pivot fitting.

3 Coat a new bolt (60) with corrosion preventive compound (Grade 2).

4 Position stay assembly (65), and install washer (64), shim (63), washer (62) (if fitted), washer (61) and bolt (60) at front and rear pivot fittings. Torque bolt (60), at each end 100 to 150 pound-inches.

5 Drill the new bolt (60) for cotter pin installation. Install a new cotter pin (59).

b) Install Platen Assembly

1 Install two screw locking inserts (56) to platen (48).

2 Position rebound rubber assembly (55) on platen (48) and secure with two washers (54), two bolts (53), bolt (52), washer (51), and nut (50).

3 Coat four screw-locking inserts (49) with corrosion protective compound (Grade 2) and install in pads on leg.

4 At pads on leg, clean threads in each pad locking insert, and threads of twelve-point bolts (46) by brushing thoroughly with DS108. Wipe dry with lint free cloth.

5 Coat all threads with Loctite 'N' 7649 Primer (ASTM D5363, Grade N), and allow to air dry for five minutes.

6 When activator is dry, apply an even coat of Loctite Grade 'H' (ASTM D5363, AN0161) sealant to all threads.

7 Position platen assembly complete with rebound assembly on pads, and install four washers (47) and four twelve-point bolts (46). Torque bolts 480 to 500 pound-inches.

8 Cure assembly at 120° - 150°F for 30 minutes under a heat lamp.

9 Position scuff plate (44) with liner (45), against face of platen assembly and install two spigots (43), two screws (42), two washers (41), and two nuts (40).

c) Install Axle

1 Apply corrosion preventive compound (Grade 2) to bolt holes in axle and leg.

2 Position axle (31) and secure to leg with bolt (30), washer (29), and a nut (28).

d) Adhesive Bond Jacking Pad to Leg Machining

WARNING **USE SOLVENTS AND ADHESIVES IN A WELL VENTILATED AREA AND KEEP AWAY FROM OPEN FLAME. AVOID TOUCHING PREPARED BOND SURFACE AREAS WITH BARE HANDS. WEAR CLEAN, SOLVENT RESISTANT GLOVES WHEN CLEANING AND HANDLING SURFACES.**

CAUTION **DO NOT USE PROTECTIVE HAND CREAM AS IT MAY CONTAMINATE CLEAN OR ADHESIVE COATED SURFACES.**

1 Clean jacking pad bond area of leg machining, jacking pad (37), and clamp (36), using a cloth dampened with DS108 or MEK, and wipe dry with a lint-free cloth before solvent evaporates.

2 Thoroughly stir adhesive EC1357 and apply a thin, even coat to bond surfaces.

3 Allow surfaces to air dry until adhesive is tacky, but does not transfer to gloved fingers when touched lightly.

4 Measure 15.00 inches from center of hole in lug toward axle.

5 Position jacking pad and clamp so that the lower face of jacking pad and clamp is coincident with the 15.00 inch mark (refer to Figure 5). Press parts together.

6 Position bracket (35) and secure jacking pad and clamp with two bolts (34), two bolts (34A), four washers (33) and four nuts (32).

7 Remove surplus adhesive using toluene or MEK.

e) Adhesive Bond Upper Clamp to Leg Machining

1 Clean bond surface areas on leg machining (above the shoulder line, short tube side), rubber sheet (39), and clamp (40), using a cloth dampened with DS108 or MEK. Wipe dry with a clean, dry lint-free cloth before solvent evaporates.

2 Measure 0.75 inch from the shoulder line on the leg machining (short tube side) toward the pivot fitting as per Figure 5.

3 Apply a thin, even coat of adhesive RTV730 to the rubber sheet and clamp.

CAUTION RTV730 ADHESIVE IS COMPLETELY SOLVENT RESISTANT. HANDLE WITH CARE AND AVOID SPILLAGE.

4 Install rubber sheet and clamp so that the bottom face is coincident with the measured 0.75 inch mark.

f) Mount Lower Rib and Adhesive Bond Seal to Rod and Leg Machining

1 Cut Celastic tape to a dimension of 1.0 x 1.0 inch.

Note Use of Celastic tape is not applicable to leg assemblies fitted for skis.

2 Clean rib (25) area to be covered with tape, using a cloth dampened with MEK and dry with a clean cloth before solvent evaporates.

3 Thoroughly soak contact side of tape with MEK and hand smooth tape on rib. Use a non-porous rubber block or pad to shape the tape to rib surface and ensure intimate contact throughout.

4 Measure 27.33 inches from center of hole in lug toward the axle (refer to Figure 5).

5 Position lower face of rib (25) coincident with the 27.33 inches measured point. Secure rib (25) to leg machining with clamp (24).

6 Temporarily install seal (26) and trim to suit leg and rib.

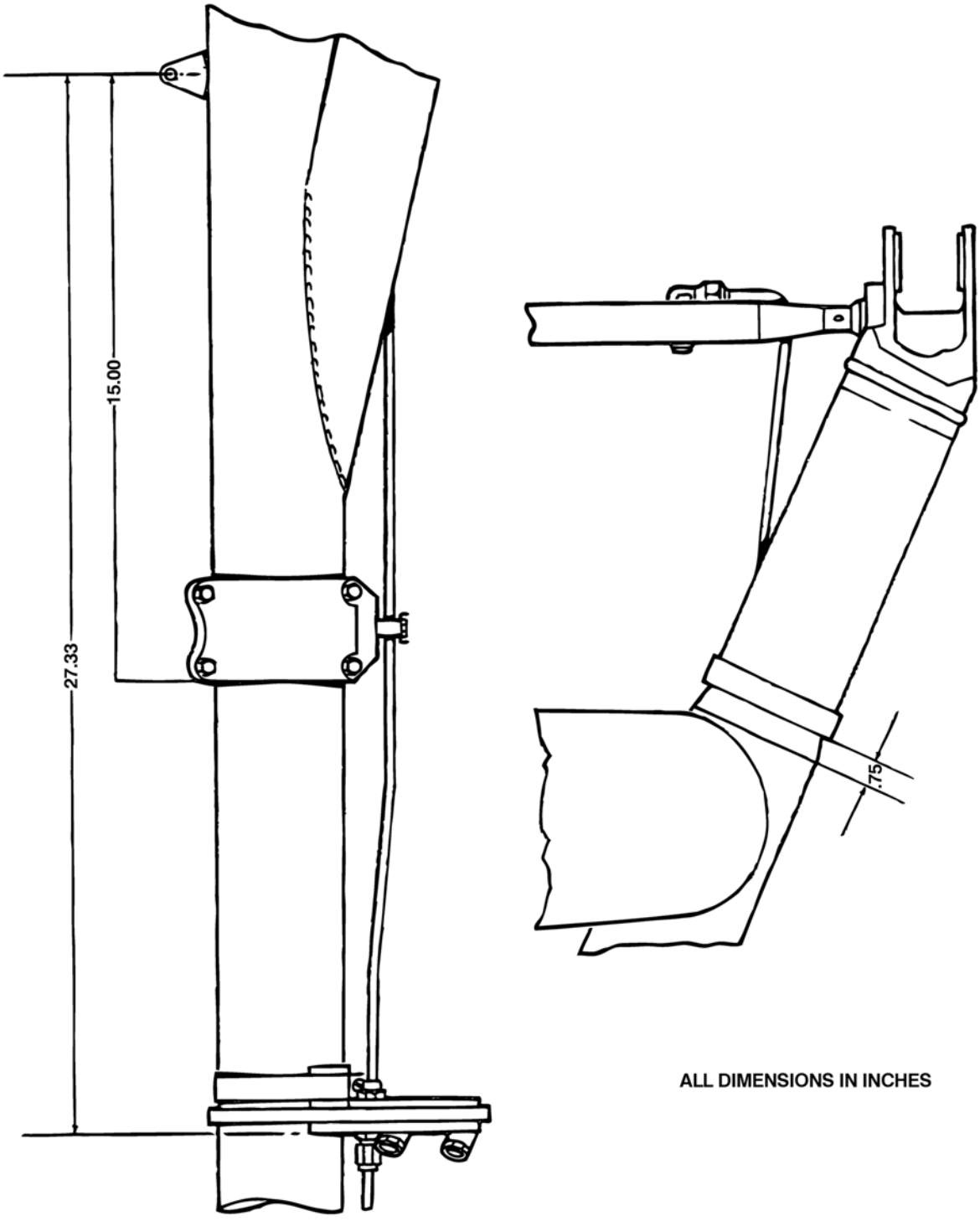
7 Clean bonding surfaces of rib, leg machining, and seal, using a cloth dampened with DS108 or MEK and wipe dry with a clean cloth before solvent evaporates.

8 Apply a thin, even coat of adhesive RTV730 to the rib and leg machining, position seal and press firmly against leg and rib.

CAUTION RTV730 ADHESIVE IS COMPLETELY SOLVENT RESISTANT. HANDLE WITH CARE AND AVOID SPILLAGE.

9 Cure adhesive bonds for 24 hours at room temperature.

WARNING ALLOW ALL BONDS TO SET BEFORE SUBJECTING ASSEMBLY TO FURTHER WORKING OR INSTALLATION.



Positioning of the Clamps for Main Landing Gear C6U1103–3/–4
Figure 5

g) Complete Rib Installation

1 Install two packings (23), two plugs (22), union (21) and nut (20) to rib.

h) Install Tube Assembly

1 Position elbow (19) on stay assembly at top of leg and secure with washer (18) and nut (17).

2 Position tube assembly (16) parallel to the leg machining.

3 Connect tube assembly to elbow (19) on stay assembly and to union (21) on rib.

Note If leg assembly is fitted for skis, install two tube assemblies and electrical wiring for ski position indication.

4 Position fairlead (15) and channel (14), and secure tube assembly to bracket (35) on jacking pad with washer (13) and bolt (12).

5 At machined lug, position angle (10) and secure with bolt (11), washer (9) and nut (8). Secure tube assembly to angle with clamp (7), screw (6), washer (5), and nut (4).

6 Secure tube assembly to clamp (38) at top of leg assembly with spacer (3), clamp (2), and nut (1).

12) Locate and record the 4 digit serial number and 2 letter suffix on the aft face of the aft pivot fitting for the inspected and repaired leg machining (C6UM1110).

13) Install main landing gear leg (refer to Para 2.B).

5. Protection for Storage

A. Preparation

1) Cap the brake tube assembly at both ends.

2) Lightly lubricate bushings with grease, conforming to MIL–PRF–23827, and wrap front and rear pivot fittings and stay assembly joints in water vapour resistant barrier material.

3) Wrap platen assembly and rebound rubber assembly in water vapour resistant barrier material.

4) Lightly lubricate axle with grease, conforming to MIL–PRF–23827, and wrap in water vapour resistant barrier material.

B. Protection for Temporary Storage (New or Newly Overhauled MLG Only)

The following steps protect and prepare the main landing gear leg machining (C6UM1110) or leg assembly for short-term storage (less than 6 months).

Note For temporary storage of up to 6 months, provision for protection is applicable only to new or newly overhauled MLG.

1) Prepare on a flat soft table.

2) Visually inspect to ensure all surfaces are appropriately plated, painted, and/or protected.

CAUTION ENSURE THAT THE SELECTED HEAT SEALING MACHINE CAN PROVIDE THE CORRECT TEMPERATURE AND PRESSURE FOR THE SELECTED FILM. FAILURE TO PROPERLY SEAL THE PACKAGE WILL COMPROMISE THE CORROSION PREVENTIVE PROPERTIES OF THE PACKAGING.

CAUTION **DO NOT PLACE MATERIALS THAT MAY CARRY OR ABSORB MOISTURE, E.G. PAPER PRODUCTS OR FOAMS, INSIDE THE PACKAGING. MOIST MATERIALS IN CONTACT WITH THE MLG MAY INITIATE CORROSION.**

3) Unroll a suitable length of multi-metal Vapor Corrosion Inhibiting (VCI) film to MIL–PRF–22019 Type I or NACE–TM0208 Grade 4 (e.g. CorTec VpCI–126), with a minimum thickness of 0.004 in (0.1 mm). Sleeving or bags made of this material can also be used.

4) Carefully place the main landing gear leg on the film or in the sleeving or bag. Use a hand-held heat sealing machine to seal the open edges, leaving a small gap on one edge. Gently press out excess air. Seal the remaining opening using the heat sealing machine. Refer to manufacturer instructions for the recommended sealing conditions (temperature, time, and pressure) for the selected film/sleeve/bag.

5) Label the exterior with the serial number, part number, overhaul date and packaging date.

6) Store the MLG indoors, away from direct sunlight. Avoid contact with moisture. Do not stack. Enclosing the MLG in a protective crate after bagging is recommended to prevent contact with other objects.

6. Troubleshooting

The DAU configuration plugs are unique to an aircraft specific gear configuration, VSP18–XXXXXX. (XXXXXX = LAND, SKIS or FLOATS). The DAU compares the configuration plug to the gear configuration data held in the MAU options file.

If the data does not match the DAU generates the following CAS Warning:

CAS Warning	Possible Cause	Check	Remedial Action
CONFIG-MISCOMPARE	Mismatch between the DAU configuration plugs and the selected MAU configurations files.	Check that the correct configuration plugs are used on DAU 1 and DAU 2. Check correct option file is selected on MAU for gear configuration.	Confirm DAU configuration plugs and MAU option file settings to match the current gear configuration.

If the configuration error continues, contact Viking Air Ltd. Technical Support.

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